# Caleb Kyei

Control & Power Systems Engineer
Erasmus Mundus DREAM Master's | Based in Eindhoven, studying in Barcelona
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## PROFILE

Power electronics and control engineer with 4.5 years at Signify R&D, including two as an engineer. Built a digital PID design tool for high-power PFC that cut control-parameter design and verification from **a week to minutes**. Now completing the Erasmus Mundus DREAM Master's, with practical work in converter control, weak-grid validation, MPC and renewable-energy systems. Seeking a thesis in control, power electronics or energy systems.

**Master's thesis availability: 1 February to 31 July 2027.**

## EDUCATION

**Erasmus Mundus DREAM | Triple Master's** 2025 – 2027
*Control and Robotics; Electrical Engineering; Energy Engineering*

- **UPC Barcelona**, Energy Engineering, Sep 2026 – Aug 2027: current semester and thesis examining university; individual converter-control project at CITCEA-UPC.
- **HTW Berlin**, Electrical Engineering, Mar – Aug 2026: Power Electronics **1.0** (German scale, 1.0 best).
- **Centrale Nantes**, Control and Robotics, Sep 2025 – Feb 2026: Dynamic Components of Power Systems **A+**.

**Fontys University of Applied Sciences** 2019 – 2023
*BSc Electrical and Electronics Engineering*, *Cum Laude* | *Eindhoven*

Thesis: Digital Control in Power Systems, **9.0/10**. Advanced Power Electronics: 9.1/10; Control Theory: 9.1/10.

## SELECTED ENGINEERING PROJECTS

**Grid-Connected Converter Control** UPC | Sep 2026 – present
*Individual project at CITCEA-UPC; in progress*

- Built an averaged two-level grid-following VSC with discrete SRF-PLL and decoupled $d$-$q$ current control; checked responses against linearised and sampled-data predictions.
- Swept short-circuit ratio and X/R to assess stability versus PLL bandwidth. Implemented DC-link voltage control, current limiting and clamping anti-windup.
- Grid-forming control of a modular multilevel converter (MMC) is a planned extension, to be completed before the thesis.

**Three-Phase Inverter & Model Predictive Control** HTW Berlin | 2026
*Two two-person projects in MATLAB/Simulink*

- Modelled a 10 kVA grid-following inverter with SRF-PLL, decoupled $d$-$q$ PI current control, SPWM and analytical $LCL$ filter design.
- Implemented a state-space MPC speed controller in an EV powertrain with PMSM field-oriented control; evaluated controller variants using parallel simulation sweeps.

**Supervisory Control of a 2 MW DFIG Wind Turbine** Centrale Nantes | 2025–2026
*Control coordination and simulation validation*

Designed a supervisory FSM coordinating MPPT and rated-power modes with anti-chattering hysteresis, rotor-side-converter torque and pitch control; validated in Simulink under turbulent wind.

## PROFESSIONAL EXPERIENCE

**Signify (formerly Philips Lighting)** Eindhoven, Netherlands
*Power electronics, digital control and embedded systems*

**Electronics Development Engineer** Aug 2023 – Aug 2025

- Developed switch-mode LED drivers, including isolated flyback auxiliary supplies.
- Worked across embedded hardware/software and driver communication interfaces, including Modbus; horticulture lighting among the applications.

**Graduation Intern: Digital Control** Feb – Jul 2023

- Built a digital PID design tool for high-power PFC using MATLAB/Simulink, Python and Excel, automating parameter generation, optimisation and verification.
- Reduced control-parameter design and verification from **a week to minutes**.

**Power Electronics Intern** Feb 2022 – Jan 2023

- Investigated SiC MOSFET power stages and embedded buck-converter switching control.

**Test Automation Intern** Feb 2021 – Jan 2022

- Automated embedded testing of emergency LED drivers in R&D.

**Robotics Team Pi (FRC)** Eindhoven | Sep 2020 – Aug 2022
*Software Designer, then Mentor*

Wrote C++ autonomous aim-assist and swerve/differential drive control; later led the team's software mentoring.

## TECHNICAL SKILLS

- **Converter control:** Grid-following VSCs, $d$-$q$ current control, SRF-PLL, IMC/PID tuning, current limiting, anti-windup, field-oriented control and $LCL$ filter design.
- **Control methods:** MPC, LQR, sliding-mode control, Kalman filtering, system identification and supervisory control (FSM/Stateflow).
- **Simulation and programming:** MATLAB, Simulink, Stateflow, LTspice, Python, C, C++, Git and GitLab CI/CD.

## RELEVANT COURSEWORK & ADDITIONAL PROJECT

**UPC, in progress:** Renewable Grid Integration; Power Quality; AI in Energy Systems; Data Science for Energy Systems.

**Centrale Nantes:** Power-system stability and inertia via DAE models; MILP/KKT scheduling optimisation; robust and optimal control.

**Solar forecasting, HTW Berlin:** Built a Python day-ahead forecast for the 50Hertz control area using ENTSO-E data; ranked **2nd of 16 on RMSE**, behind the TSO baseline.

## ADDITIONAL INFORMATION

**Languages:** English (Native), Dutch (B1), French (A2+), German (A1.2).

**Training:** Advanced Modeling & Simulation of Power Electronic Systems using PLECS (Plexim, 2023).

**Recognition:** Fontys Top Talent Scholarship (four awards); ECHO mentorship programme.